

Desigualdad de Markov

Corolario 1 (Desigualdad de Markov). Sea $X \in \mathcal{L}^1(\mathbb{P})$ una v.a.y $1 \leq p < \infty$

$$\implies \forall t > 0 : \mathbb{P}(|X| \geq t) \leq \frac{\|X\|_p^p}{t^p}.$$

Demostración: Tomamos $\varphi(t) = |t|^p$ y $A = \{|X| \geq t\}$, por la desigualdad de Chebyshev:

$$t^p \cdot \mathbb{P}(|X| \geq t) \leq \mathbb{E} [\mathbb{1}_{\{|X| \geq t\}} \cdot |X|^p] \leq \mathbb{E} [|X|^p] = \|X\|_p^p.$$

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